

PROG 6212

POE Part 1

Lecturer Reece Wanvig

API End Point Plan

Alison Susan Basford

ST10472125w

RaceDay API Endpoint Plan

API endpoint tables supporting the RaceDay Event Management System.

Authentication (Register and Login)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Create a User account with Role=Participant linked to Participant profile for Participant	Public	{ userName, email, password, firstName, lastName, dob, contactNumber, emergencyContactName, emergencyContactNumber }	201 Created – Creates the User account and linked Participant profile 400 Bad Request – Required registration information is missing or invalid 409 Conflict – The username already exists.
POST	/api/auth/login	Login –Validates Username and password against username and PasswordHash	Public	{ userName, password }	200 OK – Login succeeds and returns the authenticated user's role 401 Unauthorized – The username or password is invalid.
POST	/api/auth/logout	Logout – Ends session	Authenticated User	None	204 No Content – The authenticated session has ended
GET	/api/auth/me	Returns authenticated user's details and role. Frontend displays correct page and login status.	Authenticated User	None	200 OK – Returns the authenticated user's details and role. 401 Unauthorized – No valid authenticated session exists

Event

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	View all upcoming events	Public	None	200 OK – Returns all upcoming events
GET	/api/events/{eventId}	View individual event by id	Public	None	200 OK – Returns details of the selected event 404 Not Found – Event does not exist
POST	/api/organisers/me/events	Creates a race	Authenticated Organiser	{ raceName, raceDescription, raceDate, raceLocation, raceDistance, raceEventType, registrationOpenDate, registrationCloseDate }	201 Created – Creates and returns the new event linked to the authenticated organiser 400 Bad Request – Event details are invalid 401 Unauthorised – No valid authentication session exists 403 Forbidden – The logged in user is not an organiser

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/events/{eventId}	Update race	Authenticated Organiser	{ raceName, raceDescription, raceDate, raceLocation, raceDistance, raceEventType, registrationOpenDate, registrationCloseDate }	200 OK – Updates and returns the selected event 400 Bad Request – Updated event details are missing or invalid 401 Unauthorised – No valid authentication session exists 403 Forbidden – The logged in user is not an organiser 404 Not Found – The event does not exist
PATCH	/api/events/{eventId}	Edit race	Authenticated Organiser	{ raceName, raceDescription, raceDate, raceLocation, raceDistance, raceEventType, registrationOpenDate, registrationCloseDate }	200 OK – Edits and returns the selected event 400 Bad Request – Updated event details are missing or invalid 401 Unauthorised – No valid authentication session exists 403 Forbidden – The logged in user is not an organiser 404 Not Found – The event does not exist
DELETE	/api/events/{eventId}	Delete an existing race by ID. Removes its related categories, registrations and results	Authenticated Organiser	None	204 No Content – The selected event was deleted successfully 401 Unauthorised – No valid authentication session exists 403 Forbidden – The logged in user is not an organiser 404 Not found – The event does not exist
GET	/api/organisers/me /events	View events added by Logged in Organiser	Authenticated Organiser	None	200 OK – Returns events created by the authenticated organiser 401 Unauthorised – No valid authentication session exists 403 Forbidden – The logged in user is not an organiser

Category

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId} /categories	Shows categories for one race “U18 5km”	Public	None	200 OK – Returns all categories for the selected event 404 Not Found – The selected event does not exist
POST	/api/events/{eventId} /categories	Adds a category	Authorised Organiser	{ categoryName, ageMin, ageMax, distance }	201 Created – Creates and returns the new category for the selected event 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged-in user is not an organiser 404 Not Found – The selected event does not exist

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/categories/{categoryID}	Update an existing category to an event	Authorised Organiser	{ categoryName, ageMin, ageMax, distance }	200 OK – Updates and returns the selected category 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged-in user is not an organiser 404 Not Found – The selected category does not exist
PATCH	/api/categories/{categoryID}	Edit an existing category to an event	Authorised Organiser	{ categoryName, ageMin, ageMax, distance }	200 OK – Edits and returns the selected category 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged-in user is not an organiser 404 Not Found – The selected category does not exist
DELETE	/api/categories/{categoryID}	Deletes a category to an event	Authorised Organiser	None	204 No Content – The selected category was deleted successfully 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged-in user is not an organiser 404 Not Found – The selected category does not exist

Participant

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/participants/me	Displays participants own profile	Authenticated Participant	None	200 OK – Returns the authenticated participant's profile 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged-in user is not a participant
PUT	/api/participants/me	Updates participants own profile	Authenticated Participant	{ firstName, lastName, dob, contactNumber, emergencyContactName, emergencyContactNumber }	200 OK – Updates and returns the authenticated participant's profile 400 Bad Request – Profile information is missing or invalid 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged in user is not a participant

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PATCH	/api/participants/me	Updates participants own profile	Authenticated Participant	{ firstName, lastName, dob, contactNumber, emergencyContactName, emergencyContactNumber }	200 OK – Edit and returns the authenticated participant's profile 400 Bad Request – Profile information is missing or invalid 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged in user is not a participant

Organiser

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/organisers	Creates a User account with Role = Organiser and linked Organiser profile (Under construction)	Authorised Organiser	{ userName, email, passwordHash, firstName, lastName, contactNumber }	201 Created – Creates and returns the new organiser User account and linked Organiser profile 400 Bad Request – Organiser details are missing or invalid 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged-in user is not an organiser
GET	/api/organisers/me	Organiser can view own profile and email address	Authorised Organiser	None	200 OK – Returns the authenticated organiser's profile 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged-in user is not an organiser
PUT	/api/organisers/me	Organiser can update their own profile and linked email address	Authorised Organiser	{ firstName, lastName, contactNumber, email }	200 OK – Updates and returns the authenticated organiser's profile 400 Bad Request – Profile information is missing or invalid 401 Unauthorized – No valid authenticated session exists. 403 Forbidden – The logged-in user is not an organiser
PATCH	/api/organisers/me	Organiser can update their own profile and linked email address	Authorised Organiser	{ firstName, lastName, contactNumber, email }	200 OK – Edit and returns the authenticated organiser's profile 400 Bad Request – Profile information is missing or invalid 401 Unauthorized – No valid authenticated session exists. 403 Forbidden – The logged-in user is not an organiser

Registration

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/registrations	Enters the authenticated participant into one event and one category. Duplication prevented by Unique ParticipantIDEventID	Authenticated Participant	{ eventID, categoryID }	201 Created – Creates and returns the participant's registration for the selected event and category 400 Bad Request – The event is closed, the participant is not eligible, or the submitted details are invalid 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged in user is not a participant 409 Conflict – The participant is already registered for this event
GET	/api/registrations/me	Shows all events entered by authenticated participant	Authenticated Participant	None	200 OK – Returns all registrations for the authenticated participant 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged in user is not a participant
GET	/api/events/{eventID}/registrations	Shows all participants registered in a particular event	Authorised Organiser	None	200 OK – Returns all registrations for the selected event 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged in user is not a organiser

Result

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Creates a result for a valid registration the system calculates positioning from finishTime within the same event	Authorised Organiser	{ registrationID, finishTime }	201 Created – Creates and returns a result for the selected registration 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged in user is not an organiser 404 Not Found – The selected registration does not exist
PUT	/api/results/{resultID}	Updates the finishTime for the selected result and system recalculates positioning if needed	Authorised Organiser	{ finishTime }	200 OK – Updates and returns the selected result, then recalculates positioning 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged in user is not an organiser 404 Not Found – The selected result does not exist

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
DELETE	/api/results/{resultID}	Deletes participant result	Authorised Organiser	None	204 No Content – The selected result was deleted successfully 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged in user is not an organiser 404 Not Found – The selected result does not exist
GET	/api/participants/me/results/{eventID}	Shows participant results for one selected race	Authenticated Participant	None	200 OK – Returns the authenticated participant's result for the selected event 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged in user is not a participant 404 Not Found – The selected event or result does not exist
GET	/api/participants/me/results	Shows participant results for all races	Authenticated Participant	None	200 OK – Returns all results for the authenticated participant 401 Unauthorized – No valid authenticated session exists 403 Forbidden – The logged-in user is not a participant
GET	/api/events/{eventID}/results	View results for a particular race, no names, RaceDayNumber, FinishTime and Positioning	Public	None	200 OK – Returns the anonymised leaderboard for the selected event, including RaceDayNumber, FinishTime, and Positioning 404 Not Found – The selected event does not exist

AI Assistance

ChatGPT (OpenAI) was used as a learning proof reading and support tool during the development of this project. It was also used to neaten up documentation for a more professional look.